

2026

GROUP 5



**AN ARDUINO BASED
TURBIDITY SENSOR
THE DUCKSAVER**



THE TEAM

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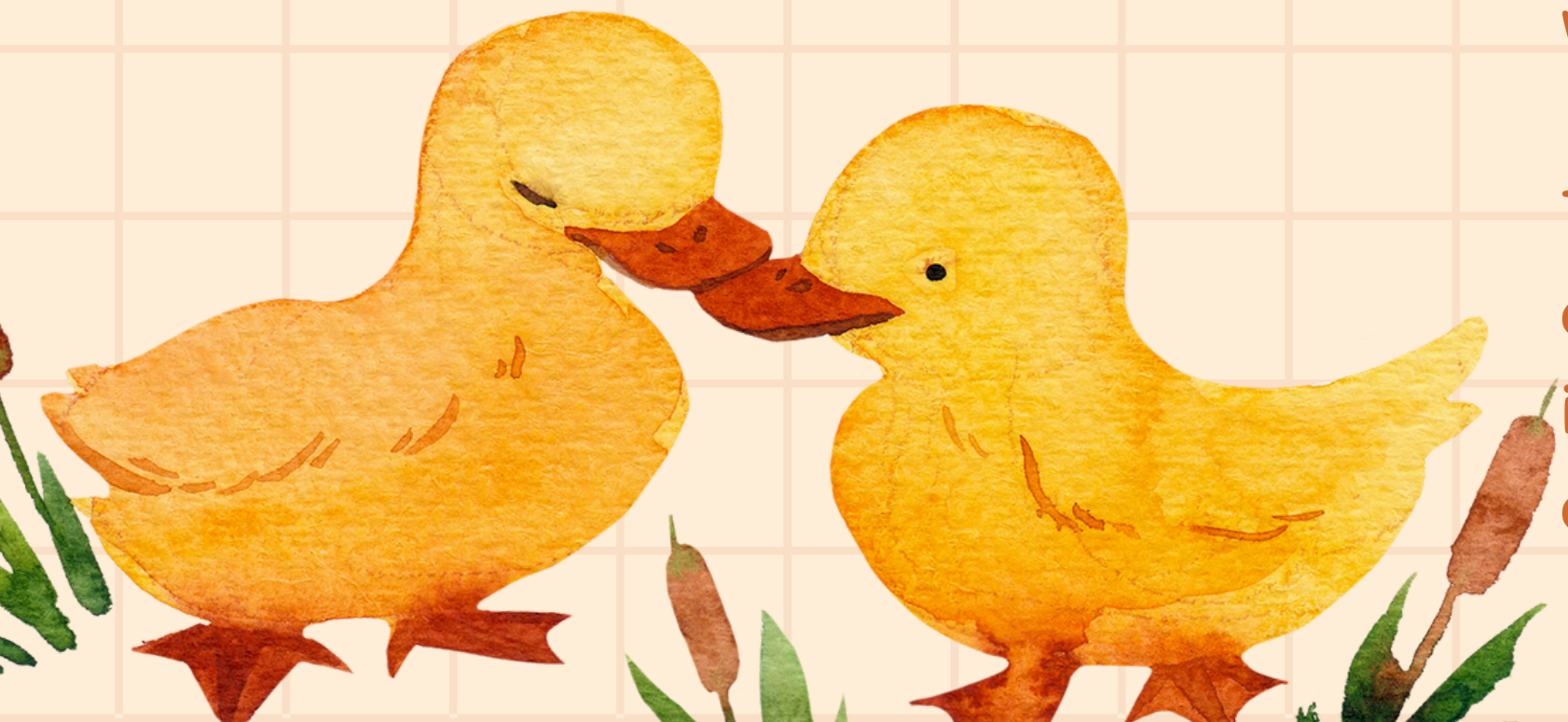
INTRODUCTION



The turbidity of water directly impacts the wildlife present in water areas. Indeed, with the decrease of sunlight, algae and other water plants will not be able to perform photosynthesis correctly. Therefore, with some species depending on these plants to feed themselves, the overall population of living beings could collapse. This means a reduction of available food for ducks.

Moreover, the turbidity will also obstruct the vision of ducks which will no longer be able to detect predators correctly.

To ensure a safe amount of turbidity, we aim to create an embedded arduino based turbidity sensor which will display in real time the turbidity level and give an alert if there is a dangerous level of turbidity.



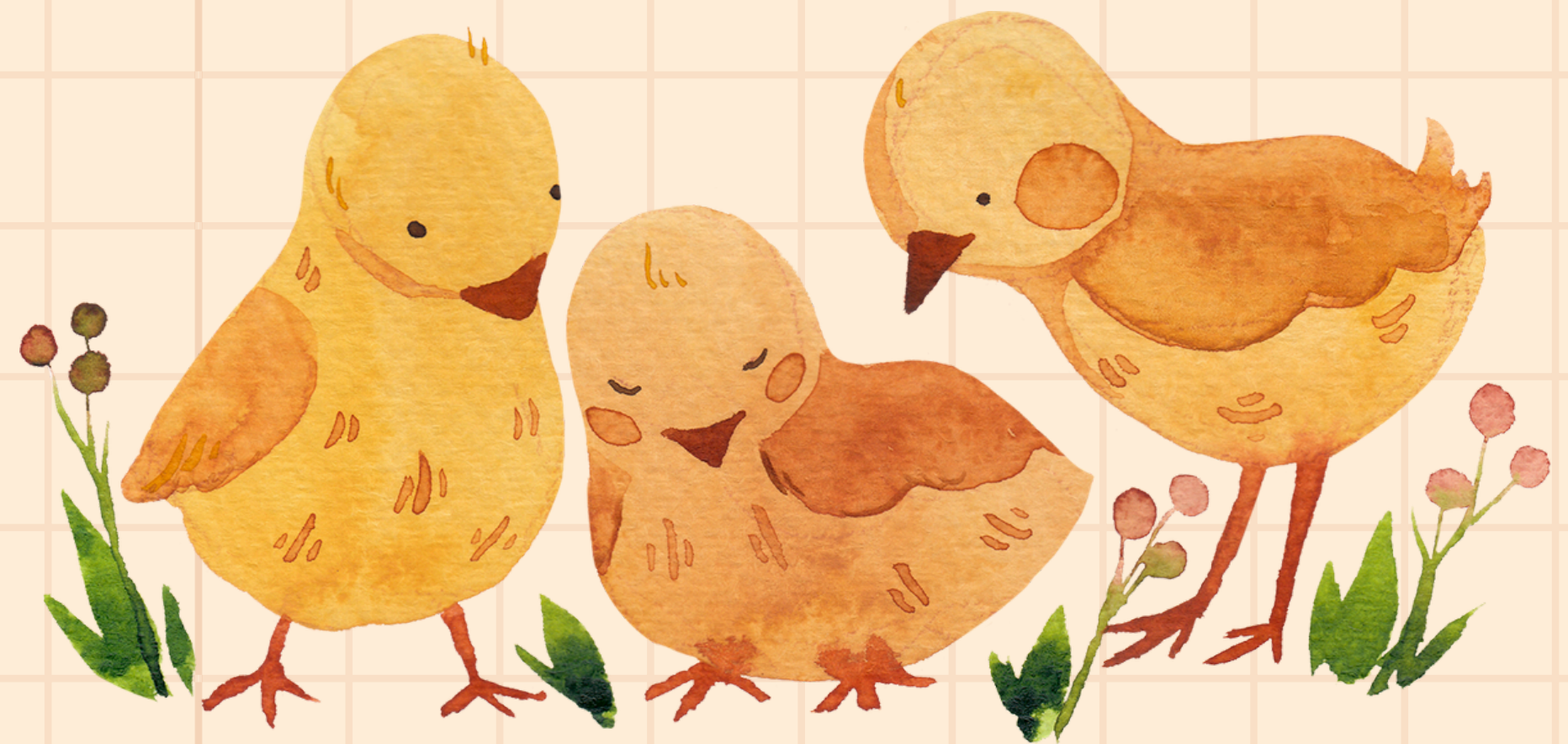


IMPLEMENTATION

The hardware used to implement the Ducksaver is composed of :

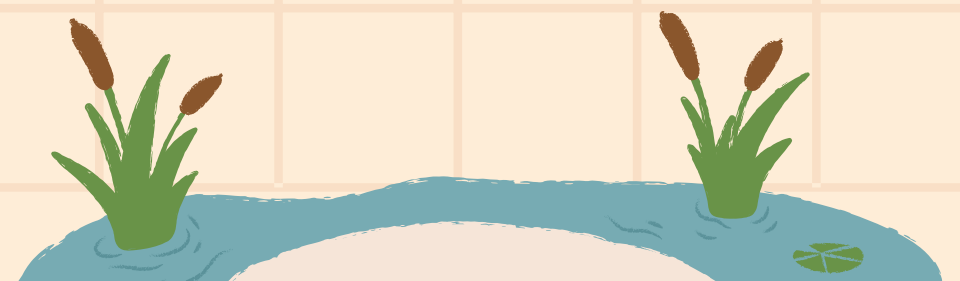
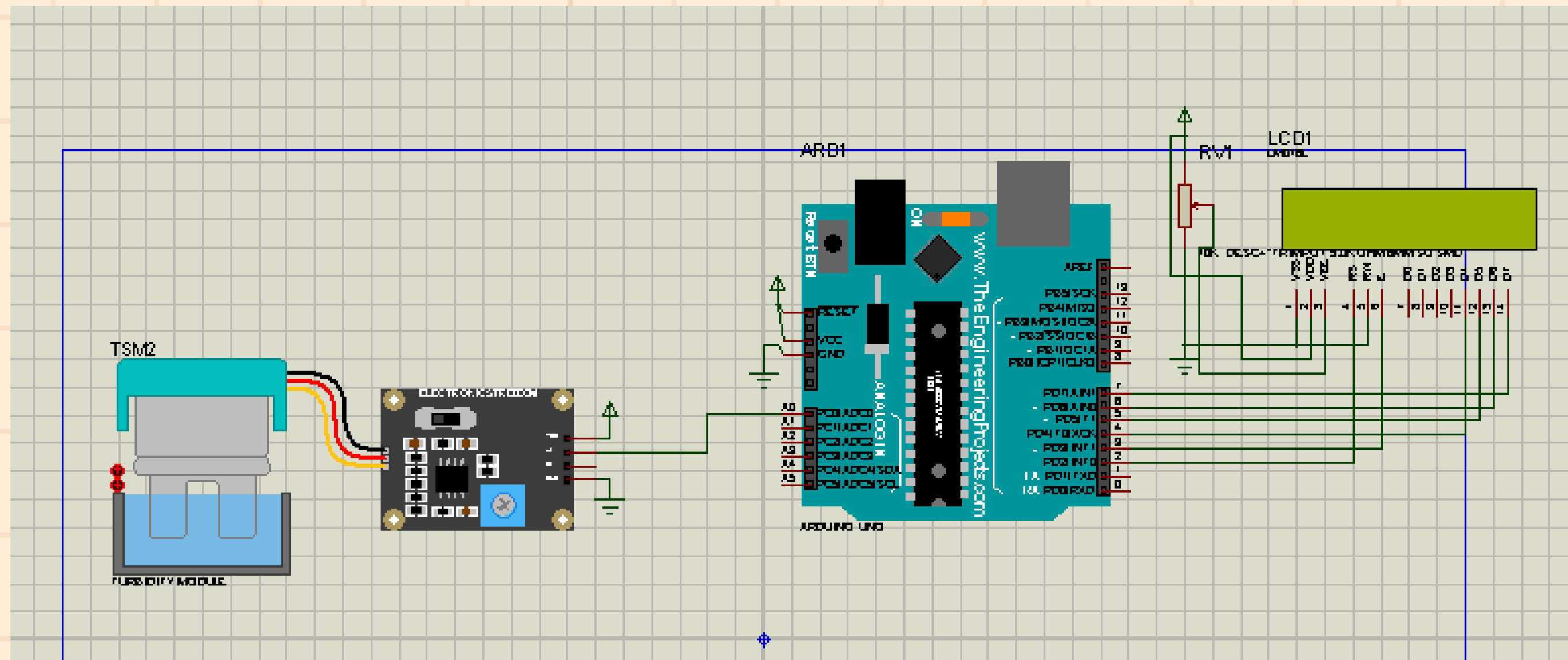
- Arduino Uno
- Breadboard / Wires
- Analog turbidity sensor + Probe
- LCD screen
- Status LEDs
- 9V Battery

The software is implemented using AVR assembly.





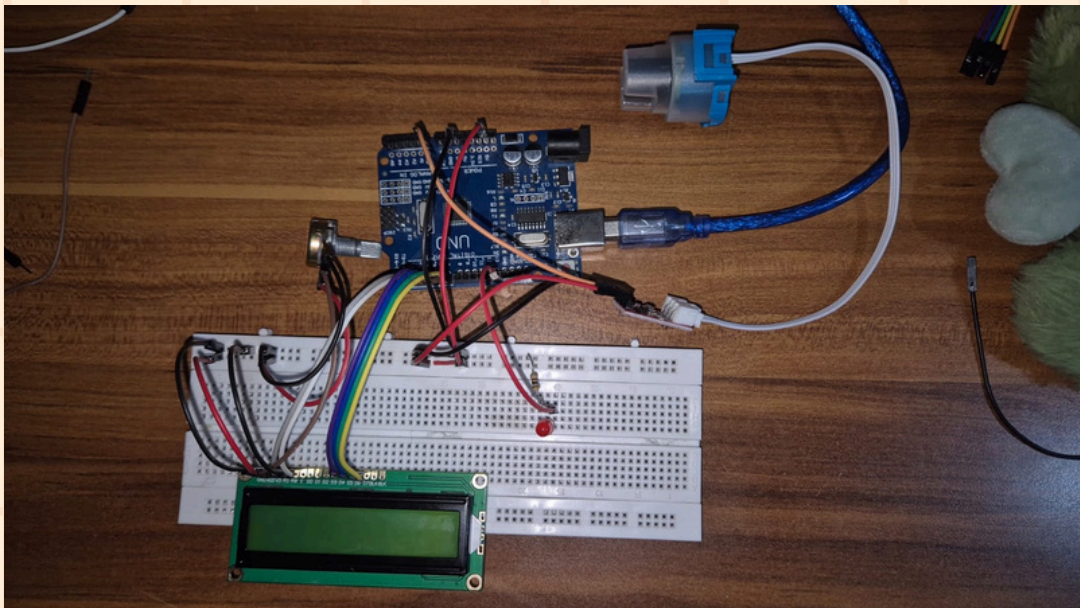
SCHEMATIC



HARDWARE

HARDWARE COMPONENTS

Component	Qty	Description
Arduino Uno (ATmega328P)	1	Core microcontroller
Analog Turbidity Sensor + Probe	1	Optical sensing module
16x2 Character LCD	1	4-bit parallel display
Status LEDs (Clean / Normal / Dirty)	3	Visual alarm indicators
9V Battery	1	Standalone power source
Breadboard & Wires	1 set	Physical connections



PIN CONFIGURATIONS

Function	ATmega328P Pin	Arduino Pin
Turbidity Sensor Input	PC0	A0
LCD Register Select (RS)	PB0	D8
LCD Enable (EN)	PB1	D9
LCD Data Bus (D4–D7)	PD4–PD7	D4, D5, D6, D7
Clean LED	PD4	D4
Normal LED	PD5	D5
Dirty LED	PB5	D13 (onboard)
UART TX	PD1	D1 / TX
UART RX	PD0	D0 / RX



SOFTWARE



- Interrupt-Driven Scheduling: Operates via Timer1 in CTC mode.
- NTU Scaling and Computation: Computes specific Nephelometric Turbidity Units (NTU), including full numerical overflow handling (>999 NTU).
- Three-Tier Water Turbidity Classification: Categorizes aquatic safety into three specific zones using conditional branching:
 - CLEAN: Low turbidity conditions.
 - NORMAL: Medium turbidity conditions.
 - DIRTY: High turbidity / contaminated conditions.
- LCD Driver: Controls a 16x2 character LCD using nibble transmission method (send_nibble) across multiple I/O ports.
- Tri-LED Visual Alarm : Dedicated digital output pins drive distinct status indicators corresponding directly to each individual water safety tier (CLEAN, NORMAL, DIRTY).
- USART communication : Outputs real-time ASCII data values (NTU: XYZ) over USART serial interface at 9600 baud rate.



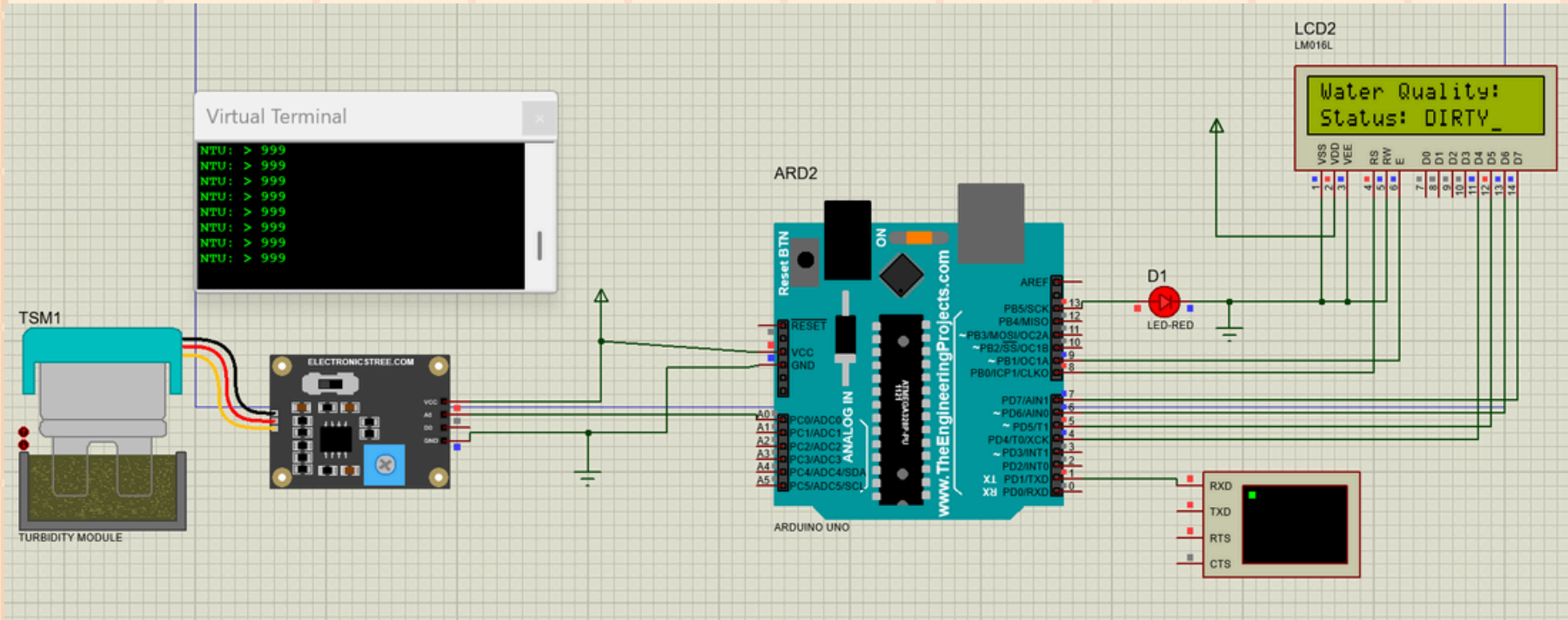


RESULTS

Test	Baseline Value (code)	NTU Reading	LCD Status	LED
Dirty Water	819	> 999	Status: DIRTY	Red ON
Normal Water	200	325	Status: NORMAL	Off
Clean Water	100	162	Status: CLEAN	Off

Parameter	Value
DIRTY threshold	$NTU \geq 500 \rightarrow$ red LED triggered
NORMAL threshold	$300 \leq NTU < 500 \rightarrow$ no LED
CLEAN threshold	$NTU < 300 \rightarrow$ no LED
Overflow handling	$NTU > 999 \rightarrow$ prints "> 999"
Update interval	Every ~2 sec via Timer1 CTC
Serial output	Confirmed working at 9600 baud

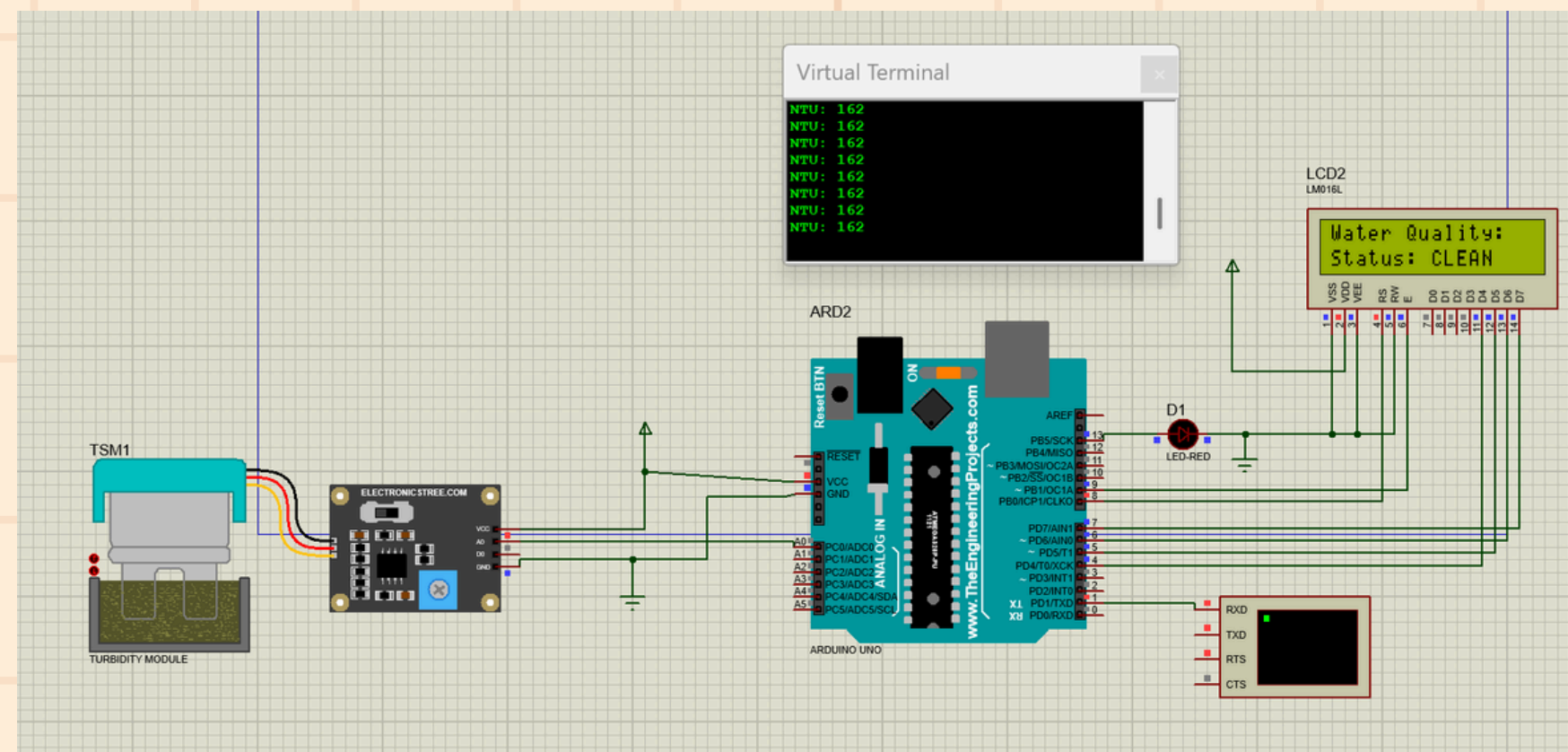
DIRTY WATER



A decorative illustration of green leaves and stems on a light orange background with a faint grid pattern. The leaves are arranged in a diagonal line from the bottom left towards the top right. The leaves are various shades of green, with some showing darker green veins. The stems are thin and green. The overall style is clean and modern.



A vibrant, stylized illustration of various flowers and greenery. The composition includes white daisies with yellow centers, orange and red tulips, and lush green leaves. The background is a light beige grid pattern.

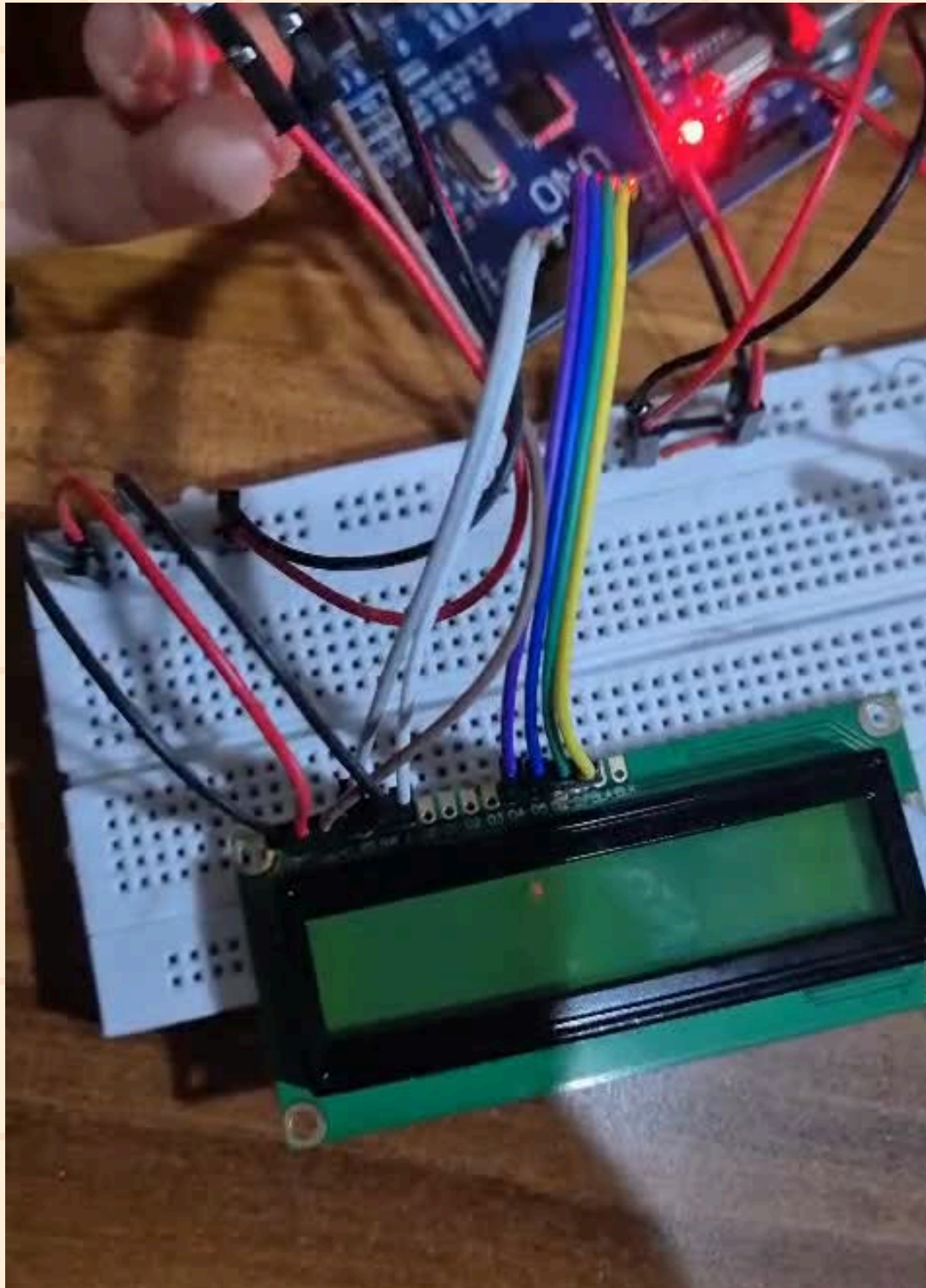




ANALYSIS

- The Ducksaver reads analog voltage from the turbidity sensor through pin A0, which the ADC converts into a digital value (0–1023). Cleaner water allows more light through, producing a higher ADC value, while dirtier water blocks light, giving a lower ADC value.
 - This ADC value is then plugged into the formula $NTU = (819 - ADC) \times 269 / 614$ to calculate turbidity. The larger the difference from baseline 819, the higher the NTU, meaning dirtier water.
 - The system then classifies the result into three tiers, CLEAN ($NTU < 300$), NORMAL ($300-499$), and DIRTY (≥ 500) and responds accordingly by updating the LCD, serial output, and red LED.
 - In simulation, all three states were successfully triggered and verified, confirming the system works correctly end-to-end.
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DRAWBACKS



During physical circuit implementation, the 16x2 LCD display persistently glitched even after being replaced twice. This suggests the issue likely stems from unstable power supply, noise on the data lines, or timing inconsistencies in the 4-bit communication rather than a faulty LCD unit.

This highlights a common gap between simulation and real world hardware, where factors like signal noise and power fluctuations are not accounted for in Proteus.

**THANK YOU FOR YOUR
ATTENTION**

